

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1166

Taloga 138 kV
September 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1166. The Interconnection Customer proposes a 103.6 MW battery energy storage facility interconnecting at the Taloga 138 kV substation in the Virginia area. The request was received on March 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 369 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$953,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: New London - Declo Substation 138 kV
Contingency: Sherburne - Bath County 345 kV
Loading: 100.7% (Rating: 425 MVA)
DFAX: 6.6%
Pre-project loading: 99.1%
2. Facility: Lowmoor - Longview2 345 kV Line
Contingency: Bath County - Longview2 345 kV L
Loading: 100.4% (Rating: 1,268 MVA)
DFAX: 20.7%
Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New London - Declo Substation 138 kV	Reconductor line	\$11,605,000	3% (\$390,000)
Lowmoor - Longview2 345 kV L	Reconductor line	\$31,732,000	2% (\$563,000)
	Total allocated to project		\$953,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.